

UNIVERSITY OF MUMBAI
Department of Information Technology

PROJECT PROPOSAL

MealMentor AI

An AI-Powered Personalized Nutrition Analysis and Diet Recommendation Android Application

Submitted in partial fulfillment of the requirements for the degree of
Bachelor of Science in Information Technology

Submitted by: Irfan Pathan

SAP ID: 53013240006

Under the Guidance of:
Dr. Prashant Chaudhary

Programme: B. Sc. Information Technology (2026–27) Semester V
Course: Field Project | Code: UIFPR310

Academic Year: 2026 – 2027

CERTIFICATE

This is to certify that the project titled “**MealMentor AI: An AI-Powered Personalized Nutrition Analysis and Diet Recommendation Android Application**” is a bonafide work carried out by **Irfan Pathan (SAP ID: 53013240006)**, a student of **B. Sc. Information Technology, Semester V**, in partial fulfillment of the requirements for the course **Field Project (UIFPR310)** during the academic year **2026–2027**.

This project has been carried out under my guidance and supervision.

Dr. Prashant Chaudhary

Project Guide

Date: _____

Head of Department

Dept. of Information Technology

Date: _____

External Examiner:

Name: _____

Signature: _____

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my project guide, **Dr. Prashant Chaudhary**, for his invaluable guidance, constant encouragement, and constructive suggestions throughout the course of this project. His expertise and mentorship have been instrumental in shaping the direction and quality of this work.

I extend my thanks to the **Head of Department** and the **faculty members** of the Department of Information Technology for providing the necessary infrastructure and academic environment to carry out this project.

I am also grateful to my family and friends for their unwavering support and motivation during the development of this project.

Finally, I would like to acknowledge the open-source community and the developers of the tools and technologies used in this project, whose contributions made this work possible.

Irfan Pathan

SAP ID: 53013240006

ABSTRACT

MealMentor AI is an intelligent Android application that leverages artificial intelligence and cloud computing to serve as a personal diet coach. The application enables users to photograph their meals, automatically identifies the food items using computer vision (Azure OpenAI GPT-4o Vision), extracts detailed nutritional information, and maintains a comprehensive daily food log. Unlike conventional calorie-tracking apps that require tedious manual entry, MealMentor AI employs image recognition to make food logging effortless and accurate.

The core differentiator of MealMentor AI is its **AI Dietitian module** — a context-aware conversational assistant that behaves like a professional dietitian. It considers the user's complete health profile (age, weight, medical conditions, dietary preferences, allergies), parsed health reports (blood work, cholesterol levels), and the specific foods consumed on that day to provide highly personalized meal recommendations. When a user asks "What should I eat for dinner?", the AI analyzes remaining caloric and macronutrient budgets, identifies nutritional gaps, and suggests meals that precisely fill those gaps while respecting all health constraints.

The application features interactive data visualization through detailed charts — daily calorie trend lines, macronutrient donut charts, weekly progress bar graphs, and nutrient radar charts — giving users clear visual insight into their dietary patterns. The complete system is built on **Microsoft Azure cloud services** (App Service, PostgreSQL, Blob Storage, OpenAI Service), ensuring scalability, reliability, and enterprise-grade security.

Keywords: Artificial Intelligence, Nutrition Analysis, Computer Vision, Natural Language Processing, Diet Recommendation, Android Application, Azure Cloud Services, Health Informatics, Personalized Healthcare

TABLE OF CONTENTS

	Title Page	i
	Certificate	ii
	Acknowledgement	iii
	Abstract	iv
	Table of Contents	v
	List of Figures	vi
	List of Tables	vii
	List of Abbreviations	viii
1	Introduction	1
1.1	Background	1
1.2	Motivation	2
1.3	Objectives	3
1.4	Scope of the Project	3
1.5	Organization of the Report	4
2	Problem Statement	5
2.1	Existing Problems	5
2.2	Proposed Solution	6
3	Literature Survey	7
3.1	Existing Systems and Applications	7
3.2	Research Papers Reviewed	8
3.3	Comparative Analysis	10
3.4	Research Gap	11
4	Proposed System	12
4.1	System Overview	12
4.2	Key Features	12

4.3	Advantages over Existing Systems	14
5	System Architecture	15
5.1	High-Level Architecture	15
5.2	Data Flow Diagram	16
5.3	Use Case Diagram	17
5.4	Sequence Diagrams	18
5.5	ER Diagram	19
6	Module Description	20
6.1	User Authentication Module	20
6.2	Profile & Health Management Module	20
6.3	Food Image Analysis Module	21
6.4	Daily Meal Tracking Module	22
6.5	Nutrition Analytics & Charts Module	22
6.6	AI Dietitian Chat Module	23
6.7	Health Report Analysis Module	24
6.8	Meal Suggestion Engine Module	24
7	Technology Stack	25
8	Database Design	28
9	System Requirements	31
10	Implementation Plan & Timeline	32
11	Expected Outcomes	33
12	Future Scope	34
13	References	35

LIST OF FIGURES

Fig 5.1	High-Level System Architecture Diagram	15
Fig 5.2	Data Flow Diagram (Level 0)	16
Fig 5.3	Data Flow Diagram (Level 1)	16
Fig 5.4	Use Case Diagram	17
Fig 5.5	Sequence Diagram — Food Image Analysis	18
Fig 5.6	Sequence Diagram — AI Dietitian Chat	18
Fig 5.7	Entity Relationship Diagram	19
Fig 6.1	Food Analysis Pipeline Flowchart	21
Fig 6.2	AI Dietitian Context Building Flow	23
Fig 10.1	Gantt Chart — Project Timeline	32

LIST OF TABLES

Table 3.1	Comparison of Existing Nutrition Apps	10
Table 7.1	Technology Stack Summary	25
Table 8.1	Users Table Schema	28
Table 8.2	Meals Table Schema	29
Table 8.3	Nutrition Entries Table Schema	29
Table 8.4	Health Reports Table Schema	29
Table 8.5	Chat Messages Table Schema	30
Table 8.6	Daily Summaries Table Schema	30
Table 9.1	Hardware Requirements	31
Table 9.2	Software Requirements	31
Table 10.1	Implementation Timeline	32

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
APK	Android Package Kit
BLOB	Binary Large Object
CRUD	Create, Read, Update, Delete
CSS	Cascading Style Sheets
DFD	Data Flow Diagram
ER	Entity Relationship
GPT	Generative Pre-trained Transformer
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
JWT	JSON Web Token
LLM	Large Language Model
ML	Machine Learning
NLP	Natural Language Processing
REST	Representational State Transfer
SDK	Software Development Kit
SQL	Structured Query Language
UI	User Interface
UML	Unified Modelling Language
UX	User Experience

CHAPTER 1

INTRODUCTION

1.1 Background

The global burden of diet-related diseases has reached unprecedented levels. According to the World Health Organization (WHO), unhealthy diets are among the top risk factors for non-communicable diseases (NCDs) including cardiovascular disease, type-2 diabetes, and certain cancers, which collectively account for approximately 71% of all deaths globally. In India specifically, the National Family Health Survey (NFHS-5, 2019–2021) reports that 24% of adults are overweight or obese, while paradoxically, 18.7% of women and 16.2% of men are underweight — highlighting a dual burden of malnutrition.

A fundamental challenge in addressing nutritional health is the difficulty individuals face in accurately tracking their food intake and understanding the nutritional implications of their dietary choices. Traditional methods of food logging — such as manually searching databases, reading nutritional labels, or maintaining handwritten food diaries — are tedious, error-prone, and unsustainable for long-term adherence. Research published in the *Journal of Medical Internet Research* indicates that manual food logging has an abandonment rate exceeding 60% within the first two weeks of use.

Recent advancements in Artificial Intelligence (AI), specifically in computer vision and Large Language Models (LLMs), have opened transformative possibilities in healthcare and nutrition science. Vision-capable AI models such as GPT-4o can now identify food items from photographs with remarkable accuracy, while conversational AI can provide context-aware, personalized dietary guidance that was previously available only through expensive consultations with professional dietitians.

MealMentor AI is conceived at the intersection of these technological capabilities and pressing public health needs. It is an intelligent Android application that combines AI-powered food image recognition, automated nutritional analysis, comprehensive meal tracking, interactive data visualization, and a conversational AI dietitian — all integrated into a seamless mobile experience accessible to anyone with a smartphone.

1.2 Motivation

The motivation behind developing MealMentor AI stems from the following observations:

1. **Accessibility Gap:** Professional dietitian consultations cost ₹500–₹2,000 per session in India, making personalized nutrition guidance inaccessible to a large portion of the population. An AI-powered mobile application can democratize access to dietary guidance.
2. **Manual Logging Fatigue:** Existing calorie-tracking apps (e.g., MyFitnessPal, HealthifyMe) rely heavily on manual food entry from databases. Users frequently

abandon these apps due to the tedious nature of searching and logging each food item. AI-powered image recognition can reduce this friction dramatically.

3. **Lack of Contextual Intelligence:** Most nutrition apps provide generic recommendations. They do not consider the user's complete health context — their medical conditions, blood work results, what they've already eaten that day, and their specific nutritional gaps at that moment. MealMentor AI addresses this by building a comprehensive health profile and providing real-time, context-aware recommendations.
4. **Cloud Computing Evolution:** Microsoft Azure's ecosystem now offers AI, database, storage, and compute services that can be integrated to build sophisticated health applications without requiring extensive infrastructure expertise — making it ideal for a final year project that demonstrates modern software engineering practices.
5. **Personal Health Awareness:** The post-COVID era has seen a significant increase in health consciousness among young adults. There is a growing demand for intelligent health tools that go beyond simple calorie counting.

1.3 Objectives

The primary objectives of this project are:

1. To design and develop an Android application that uses AI-powered computer vision (Azure OpenAI GPT-4o) to identify food items from photographs and automatically extract detailed nutritional information.
2. To implement a comprehensive daily meal tracking system that maintains a persistent log of all foods consumed, enabling users to monitor their caloric and macronutrient intake against personalized targets.
3. To build an AI Dietitian chat module — a context-aware conversational assistant that provides personalized meal recommendations based on the user's health profile, medical conditions, dietary preferences, and the specific foods consumed on the current day.
4. To create interactive and detailed data visualizations (line charts, donut charts, bar graphs, radar charts) that present nutritional trends over daily, weekly, and monthly periods.
5. To implement a health report analysis feature that allows users to upload medical reports (blood work, general checkups) for AI-based parsing, integrating the extracted health markers into dietary recommendations.
6. To deploy the backend infrastructure on Microsoft Azure cloud services (App Service, Database for PostgreSQL, Blob Storage, OpenAI Service), demonstrating proficiency in cloud-native application development.
7. To deliver a polished, production-quality Android application (APK) with a premium user interface and smooth user experience.

1.4 Scope of the Project

In Scope:

- Android mobile application development using React Native (Expo)
- AI-powered food image recognition and nutritional analysis
- User registration, authentication, and profile management
- Daily meal logging (both AI-assisted and manual entry)
- Interactive nutritional charts and analytics dashboard
- Conversational AI Dietitian with context-aware recommendations
- Health report upload and AI-based parsing
- Cloud deployment on Microsoft Azure
- RESTful API backend with Node.js and Express

Out of Scope:

- iOS application (can be extended as future work since React Native supports cross-platform)
- Real-time integration with wearable devices (smartwatches, fitness bands)
- Integration with external electronic health record (EHR) systems
- Barcode/QR code scanning for packaged foods
- Social features (sharing meals, community forums)
- Multi-language support (English only in current version)

1.5 Organization of the Report

This report is organized into the following chapters:

- **Chapter 1 — Introduction:** Provides the background, motivation, objectives, and scope of the project.
- **Chapter 2 — Problem Statement:** Formally defines the problem being addressed and the proposed solution approach.
- **Chapter 3 — Literature Survey:** Reviews existing systems, research papers, and identifies research gaps.
- **Chapter 4 — Proposed System:** Describes the MealMentor AI system, its features, and advantages.
- **Chapter 5 — System Architecture:** Presents architecture diagrams, DFDs, use case diagrams, sequence diagrams, and ER diagrams.
- **Chapter 6 — Module Description:** Details each functional module of the application.
- **Chapter 7 — Technology Stack:** Documents all technologies, frameworks, and tools used.
- **Chapter 8 — Database Design:** Presents the complete database schema and relationships.
- **Chapter 9 — System Requirements:** Lists hardware and software requirements.

- **Chapter 10 — Implementation Plan & Timeline:** Provides the phased development schedule.
- **Chapter 11 — Expected Outcomes:** Describes the anticipated results and deliverables.
- **Chapter 12 — Future Scope:** Outlines potential enhancements and extensions.
- **Chapter 13 — References:** Lists all cited works and resources.

CHAPTER 2

PROBLEM STATEMENT

2.1 Existing Problems

The current landscape of dietary management presents several critical challenges:

P1: Inaccurate Self-Reporting of Food Intake

Studies published in the *British Medical Journal (BMJ)* reveal that individuals consistently underreport their caloric intake by 30–50% when using manual food logging methods. This inaccuracy undermines any dietary plan and renders most food tracking efforts ineffective.

P2: High Abandonment Rates of Nutrition Apps

Research by Statista (2024) shows that nutrition and diet tracking apps have an average 30-day retention rate of only 14.3%. The primary reason cited is the cumbersome nature of manually searching and logging each food item consumed — a process that can take 5–10 minutes per meal.

P3: Generic, Non-Personalized Recommendations

Most nutrition applications provide standardized dietary advice based on broad demographic categories (age, gender, activity level). They fail to account for individual health conditions (diabetes, hypertension, PCOS, thyroid disorders), specific blood work results, allergies, cultural food preferences, and the dynamic nutritional context of what has already been consumed that day.

P4: Disconnected Health Data

Users' health reports (blood work, medical checkups) exist in isolation from their dietary tracking. There is no automated mechanism to connect clinical health markers (e.g., high LDL cholesterol, low Vitamin D, elevated HbA1c) with dietary recommendations in real-time.

P5: Expensive Professional Guidance

Personalized nutrition advice from certified dietitians remains financially inaccessible for the majority of the population, with consultation fees ranging from ₹500 to ₹2,000+ per session in India, and substantially higher in Western countries.

2.2 Proposed Solution

MealMentor AI addresses each of the identified problems through an integrated AI-powered mobile application:

Problem	MealMentor AI Solution
P1: Inaccurate self-reporting	AI-powered food image recognition eliminates manual guesswork; GPT-4o Vision identifies food items and portions from photographs
P2: High abandonment rates	One-tap meal logging via camera reduces friction; engaging charts and AI conversations maintain user interest
P3: Generic recommendations	Context-aware AI Dietitian considers complete health profile + today's meals + nutritional gaps to provide hyper-personalized advice
P4: Disconnected health data	Health report upload feature parses blood work and integrates findings directly into dietary recommendations
P5: Expensive professional guidance	AI Dietitian provides 24/7 personalized nutrition guidance at no per-consultation cost

CHAPTER 3

LITERATURE SURVEY

3.1 Existing Systems and Applications

3.1.1 MyFitnessPal (Under Armour, 2005)

MyFitnessPal is one of the most widely used calorie-tracking applications with over 200 million users. It maintains a database of 14+ million food items and allows users to log meals by searching the database or scanning barcodes. However, it relies entirely on manual entry, lacks AI-powered food recognition, and provides generic nutritional targets without contextual intelligence.

3.1.2 HealthifyMe (HealthifyMe Pvt. Ltd., 2012)

An India-focused nutrition and fitness platform that offers calorie tracking, diet plans, and human dietitian consultations (premium tier). Its “Snap” feature uses basic image recognition for food logging, but it does not offer a conversational AI dietitian and the premium features require paid subscription (₹999–₹4,999/month).

3.1.3 Lose It! (FitNow Inc., 2008)

Offers a food recognition feature called “Snap It” that uses computer vision to identify food from photos. While innovative, it supports a limited number of food categories, does not provide context-aware dietary recommendations, and lacks integration with health reports.

3.1.4 Nutrino (Nutrino Health Ltd., 2012 — acquired by Medtronic)

A nutrition AI platform that provides personalized meal recommendations based on health conditions, particularly diabetes. While conceptually closest to MealMentor AI’s vision, it operates as a B2B platform integrated into medical devices, not as a consumer-facing mobile application.

3.1.5 Yazio (YAZIO GmbH, 2013)

A European calorie counter and intermittent fasting tracker with meal planning features. Offers barcode scanning and manual logging but does not include AI-powered food recognition or conversational dietary guidance.

3.2 Research Papers Reviewed

Paper 1

Mezgec, S., & Koroušić Seljak, B. (2017). “NutriNet: A Deep Learning Food and Drink Image Recognition System for Dietary Assessment.” *Nutrients*, 9(7), 657. — Proposes a deep learning model using CNNs for food image recognition across 520 food categories. Achieved 86.72% top-5 accuracy. Validates the viability of AI-based food recognition. MealMentor AI

leverages more advanced vision models (GPT-4o) that achieve higher accuracy without requiring custom training data.

Paper 2

Bossard, L., Guillaumin, M., & Van Gool, L. (2014). “Food-101 – Mining Discriminative Components with Random Forests.” *European Conference on Computer Vision (ECCV)*. — Introduces the Food-101 dataset with 101,000 images across 101 food categories. Establishes benchmarks for food recognition models. GPT-4o’s training includes this and significantly larger datasets.

Paper 3

Sahoo, D., Hao, W., et al. (2019). “FoodAI: Food Image Recognition via Deep Learning for Smart Food Logging.” *KDD 2019*. — Proposes a multi-task learning framework for simultaneous food recognition, ingredient detection, and nutritional estimation from images. Demonstrates the pipeline approach that MealMentor AI implements using GPT-4o Vision.

Paper 4

Chen, J., Zhu, B., et al. (2020). “A Study of Multi-Task and Region-Wise Deep Learning for Food Ingredient Recognition.” *IEEE Transactions on Image Processing*, 30, 1514–1526. — Multi-region analysis of food images for ingredient-level recognition to improve nutritional estimation accuracy.

Paper 5

Vaswani, A., et al. (2017). “Attention Is All You Need.” *Advances in Neural Information Processing Systems (NeurIPS)*. — Introduces the Transformer architecture that forms the foundation of GPT-4o and all modern LLMs used in this project.

Paper 6

Achiam, J., et al. (2023). “GPT-4 Technical Report.” *OpenAI*. — Details the capabilities of GPT-4, including multimodal (vision) capabilities for understanding images alongside text. MealMentor AI utilizes GPT-4o (optimized version) for food image analysis and conversational diet guidance.

3.3 Comparative Analysis

Table 3.1: Comparison of Existing Nutrition Applications

Feature	MyFitnessPal	HealthifyMe	Lose It!	Yazio	MealMentor AI
AI Food Image Recognition	✗	Basic	Limited	✗	✓ GPT-4o
Manual Food Logging	✓	✓	✓	✓	✓
Barcode Scanning	✓	✓	✓	✓	✗
Detailed Nutrition Charts	Basic	Basic	Basic	✓	✓ Advanced
Conversational AI Dietitian	✗	✗	✗	✗	✓
Context-Aware Recommendations	✗	Limited	✗	✗	✓
Health Report Integration	✗	✗	✗	✗	✓
Micronutrient Tracking	Premium	Limited	Premium	Premium	✓ Full
Free Core Features	Freemium	Freemium	Freemium	Freemium	✓ Full
Cloud Platform	AWS	AWS	AWS	GCP	Azure
Open Source	✗	✗	✗	✗	✓

3.4 Research Gap

The literature survey reveals the following gaps that MealMentor AI addresses:

1. **No existing consumer app integrates GPT-4o-level vision models with a conversational AI dietitian** — existing apps either have basic image recognition OR human dietitians (at premium cost), but not an AI that combines both capabilities.
2. **Health report data remains siloed from dietary recommendations** — no reviewed system automatically parses medical reports and factors clinical markers into daily meal suggestions.
3. **Context-awareness at meal-level granularity is absent** — no existing app dynamically adjusts recommendations based on the specific foods consumed earlier the same day, calculating real-time nutritional gaps and filling them with the next meal suggestion.

CHAPTER 4

PROPOSED SYSTEM

4.1 System Overview

MealMentor AI is a three-tier Android application consisting of:

1. **Presentation Tier (Android App):** A React Native mobile application providing the user interface across 10+ screens, built with Expo for streamlined development and deployment.
2. **Application Tier (Backend API):** A Node.js + Express RESTful API server deployed on Azure App Service, handling business logic, AI orchestration, and data management through 15+ endpoints.
3. **Data Tier (Azure Cloud Services):** Azure Database for PostgreSQL (with pgvector extension for AI embeddings), Azure Blob Storage (for food images and health reports), and Azure OpenAI Service (GPT-4o for vision and chat, text-embedding-ada-002 for vector embeddings).

4.2 Key Features

Feature 1: AI-Powered Food Recognition

Users capture a photograph of their meal using the phone camera or select an image from the gallery. The image is sent to Azure OpenAI's GPT-4o Vision model, which identifies all visible food items, estimates portion sizes, and generates detailed nutritional information including calories, macronutrients (protein, carbohydrates, fats), and micronutrients (vitamins, minerals, fiber, sodium, cholesterol).

Feature 2: Comprehensive Daily Meal Tracking

Every analyzed meal is logged into the user's daily food journal with timestamp, meal type (breakfast, lunch, dinner, snack), nutritional data, and the original photograph. The dashboard displays real-time progress toward daily caloric and macronutrient targets using intuitive circular progress indicators and summary cards.

Feature 3: Interactive Nutrition Charts

The application provides four types of interactive charts:

- **Calorie Trend Line Chart:** Plots daily caloric intake over weeks/months with target line overlay
- **Macronutrient Donut Chart:** Visual split of protein, carbohydrates, and fat consumption
- **Weekly Progress Bar Chart:** Stacked bars comparing intake vs. targets across the week

- **Nutrient Radar Chart:** Spider/radar chart showing adequacy across 8+ nutrients

Feature 4: AI Dietitian Chat

A conversational interface where users can ask nutrition-related questions and receive personalized answers. The AI Dietitian has access to the complete user health profile, parsed health report data, today's complete food log with nutritional totals, calculated nutritional gaps, and 7-day meal history for pattern recognition.

Example: *"I had poha for breakfast and chole chawal for lunch. What should I have for dinner?"* → The AI calculates remaining caloric budget (~600 cal), identifies low protein intake (only 35g of 80g target), and suggests high-protein dinner options that respect the user's vegetarian preference and lactose intolerance.

Feature 5: Health Report Integration

Users can upload medical reports (blood work, general checkup results). The AI parses key health markers such as cholesterol levels (LDL, HDL, Total), blood sugar (Fasting glucose, HbA1c), vitamin levels (D, B12, Iron), kidney function markers, and thyroid levels. These parsed markers are stored and automatically factored into all future dietary recommendations.

Feature 6: Proactive Meal Suggestions

Based on the time of day and nutritional gaps, the app proactively suggests 3 meal options that would optimally fill the user's remaining nutritional needs, each with estimated nutritional breakdown and preparation difficulty.

Feature 7: Meal History & Calendar View

A searchable, filterable calendar view of all past meals with complete nutritional data, allowing users to revisit any day's eating pattern and track long-term adherence to dietary goals.

4.3 Advantages over Existing Systems

1. **Zero Manual Entry Required:** AI handles food identification and nutrition extraction automatically from photographs.
2. **True Personalization:** Recommendations consider the complete health context, not just age and weight.
3. **Dynamic, Real-Time Guidance:** Suggestions change throughout the day as meals are logged.
4. **Clinical-Grade Awareness:** Integration of blood work and health reports into dietary logic.
5. **Professional-Quality Advice at Zero Marginal Cost:** AI Dietitian available 24/7 without per-consultation charges.
6. **Rich Visual Analytics:** Charts and graphs that make nutritional data intuitive and actionable.
7. **Azure Cloud Native:** Enterprise-grade infrastructure, scalability, and data security.

CHAPTER 5

SYSTEM ARCHITECTURE

5.1 High-Level Architecture Diagram

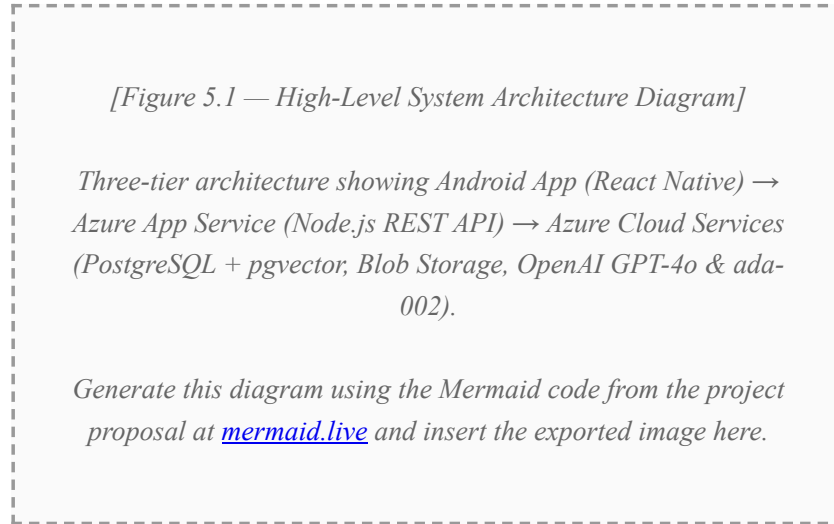


Figure 5.1: High-Level System Architecture of MealMentor AI

5.2 Data Flow Diagram

Level 0 (Context Diagram)

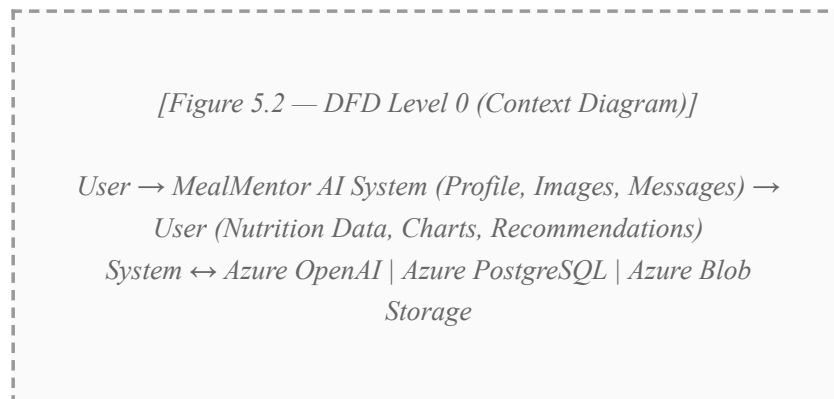


Figure 5.2: Data Flow Diagram — Level 0 (Context Diagram)

Level 1 (Expanded)

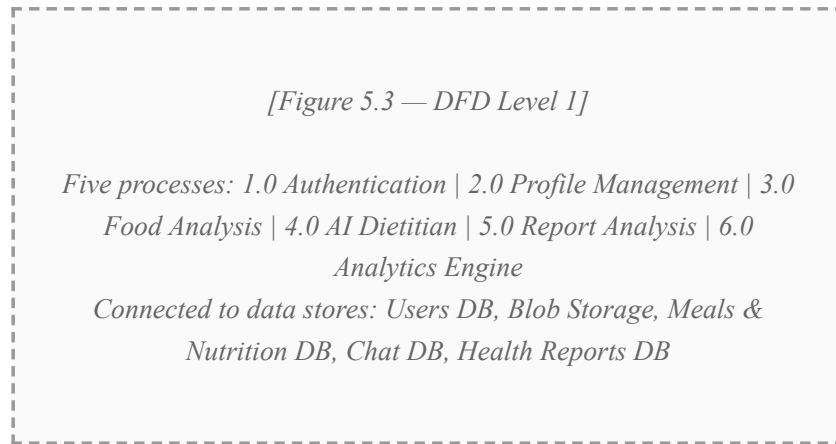


Figure 5.3: Data Flow Diagram — Level 1

5.3 Use Case Diagram

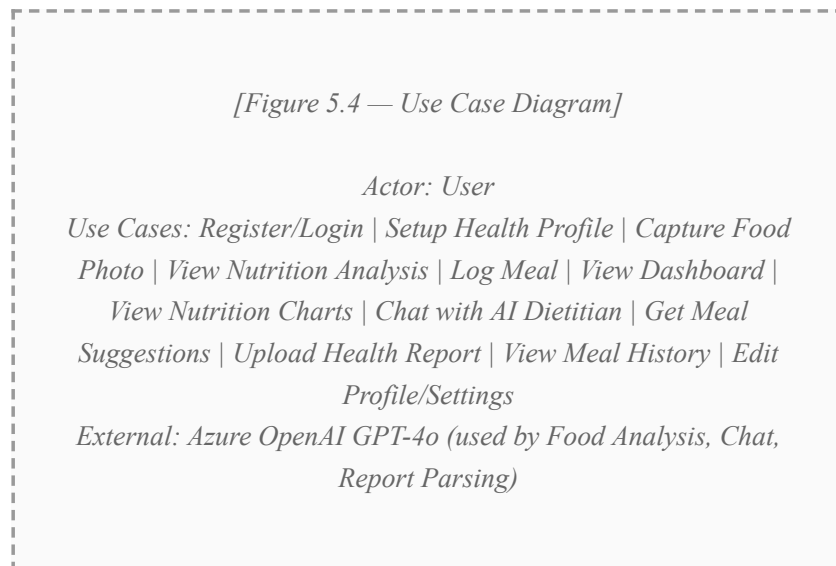


Figure 5.4: Use Case Diagram

5.4 Sequence Diagrams

Food Image Analysis Flow

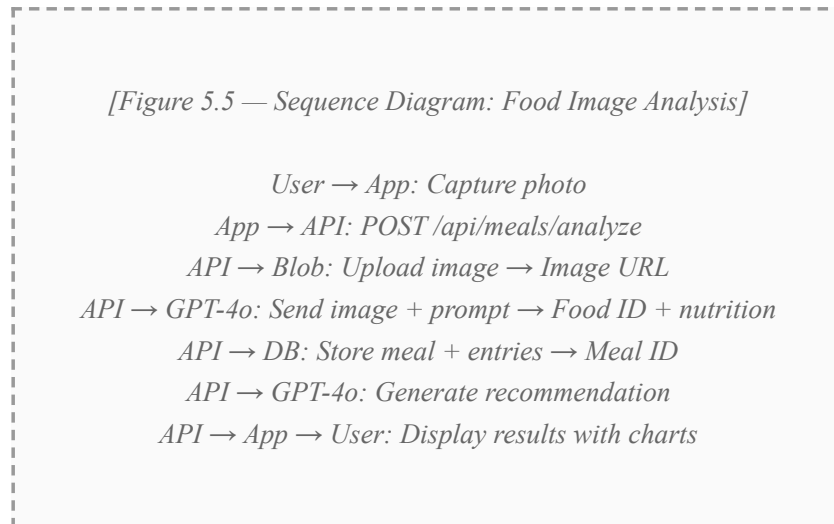


Figure 5.5: Sequence Diagram — Food Image Analysis Flow

AI Dietitian Chat Flow

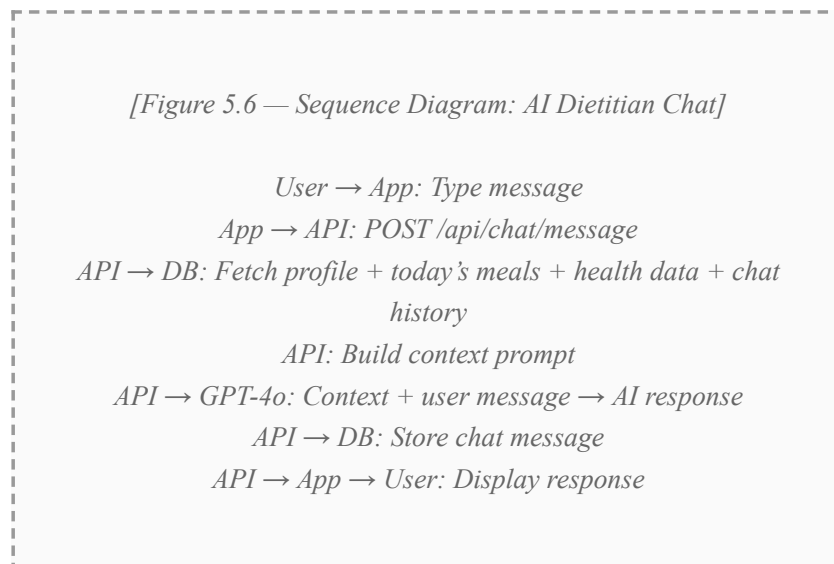


Figure 5.6: Sequence Diagram — AI Dietitian Chat Flow

5.5 ER Diagram

[Figure 5.7 — Entity Relationship Diagram]

Entities: USERS (1) → () MEALS → (*)*

NUTRITION_ENTRIES

USERS (1) → () HEALTH_REPORTS*

USERS (1) → () CHAT_MESSAGES*

USERS (1) → () DAILY_SUMMARIES*

6 tables with full attributes as specified in Chapter 8

Figure 5.7: Entity Relationship Diagram

CHAPTER 6

MODULE DESCRIPTION

6.1 User Authentication Module

Purpose: Manages user registration, login, and session management.

Functionality:

- **Registration:** Collects email and password, validates input, hashes password using bcrypt, creates user record in PostgreSQL, and returns a JWT token.
- **Login:** Validates credentials against stored hash, generates JWT token with 7-day expiry.
- **JWT Middleware:** Verifies JWT token on every authenticated request, extracts user ID for downstream operations.

Method	Endpoint	Input	Output
POST	/api/auth/register	email, password, name	JWT token, userId
POST	/api/auth/login	email, password	JWT token, userId

6.2 Profile & Health Management Module

Purpose: Manages the user's health profile, dietary preferences, and health targets.

Functionality:

- **Profile Setup:** Captures comprehensive health data — age, gender, height, weight, activity level, goal (lose/gain/maintain), caloric target, macronutrient targets, dietary preferences, allergies, and existing health conditions.
- **Auto-Calculated Targets:** If the user doesn't specify targets, the system calculates them using the Mifflin-St Jeor equation for BMR, adjusted by activity level and goal.
- **Profile Embedding:** Generates a vector embedding of the user profile using Azure OpenAI ada-002 for cosine similarity matching with food items.
- **Profile Update:** Allows users to modify any profile field, triggering re-computation of targets and embeddings.

Method	Endpoint	Input	Output
GET	/api/user/profile	JWT token	Full profile object
PUT	/api/user/profile	Updated fields	Updated profile
POST	/api/user/health-report	Report file (PDF/image)	Parsed health markers

6.3 Food Image Analysis Module

Purpose: Core module that accepts food photographs and returns detailed nutritional analysis.

Processing Pipeline:

```
User Photo → Image Optimization (resize, compress)
→ Upload to Azure Blob Storage
→ Send to GPT-4o Vision with analysis prompt
→ GPT-4o identifies food items + estimates portions
→ GPT-4o generates detailed nutrition data
→ Parse structured nutrition response
→ Generate food embedding (ada-002)
→ Store in meals + nutrition_entries tables
→ Calculate cosine similarity with user embedding
→ Generate personalized recommendation
→ Return complete analysis to app
```

GPT-4o Vision Prompt Design:

```
Analyze this food image. For each identifiable food item, provide:
1. Food item name
2. Estimated serving size
3. Detailed nutrition per serving:
  - Calories (kcal), Protein (g), Carbohydrates (g)
  - Total Fat (g), Saturated Fat (g)
  - Fiber (g), Sugar (g)
  - Sodium (mg), Cholesterol (mg)
  - Key vitamins (A, C, D, B12) as % daily value
  - Key minerals (Iron, Calcium, Potassium) as % daily value

Return the data in strict JSON format.
```

6.4 Daily Meal Tracking Module

Purpose: Maintains a comprehensive daily food log and tracks progress against targets.

- **Today's Log:** Fetches all meals logged by the user today, sorted by time.
- **Daily Totals:** Aggregates nutritional values across all meals for the day.
- **Progress Calculation:** Computes percentage of daily targets achieved for calories, protein, carbs, and fat.
- **Meal History:** Paginated retrieval of past meals with date-range filtering.
- **Manual Entry:** Allows users to search common foods and log them manually with standard nutritional values.

6.5 Nutrition Analytics & Charts Module

Purpose: Transforms raw nutritional data into interactive visual charts.

Chart Types:

1. **Daily Calorie Progress (Circular/Ring Chart):** Current calories consumed vs. target. Color-coded: green (on track), yellow (approaching limit), red (exceeded).
2. **Macronutrient Donut Chart:** Protein, carbohydrates, fat as segments. Actual values vs. target percentages.
3. **Weekly Calorie Trend (Line Chart):** 7-day calorie intake trend with target line overlay. Touch-to-reveal daily values.
4. **Monthly Nutrient Radar Chart:** 8 axes: Calories, Protein, Carbs, Fat, Fiber, Vitamins, Minerals, Sodium. Shows nutrient adequacy as a filled polygon.
5. **Weekly Macro Progress (Stacked Bar Chart):** Each day as a stacked bar showing protein/carb/fat breakdown. Target overlay for comparison.

6.6 AI Dietitian Chat Module

Purpose: The flagship feature — a conversational AI that acts as a personal dietitian.

Context Building Algorithm:

1. Fetch user profile (age, weight, conditions, preferences, allergies)
2. Fetch parsed health report data (blood markers)
3. Fetch today's complete meal log with nutritional totals
4. Calculate remaining nutritional budget:
 - Remaining calories = target - consumed
 - Remaining protein = target - consumed
 - (same for carbs, fat, fiber)
5. Identify nutritional gaps (nutrients below 50% of daily value)
6. Fetch last 10 chat messages for conversation continuity
7. Determine current time → expected meal type
8. Fetch 7-day meal pattern for preference learning
9. Compile all into structured system prompt
10. Send user message + system prompt to GPT-4o
11. Return response and store in chat history

[Figure 6.2 — AI Dietitian Context Building Flow]

Visual flowchart of the 11-step context building algorithm above.

Figure 6.2: AI Dietitian Context Building Flow

6.7 Health Report Analysis Module

Purpose: Parses uploaded health reports and extracts key clinical markers.

Processing Flow:

1. User uploads PDF/image of health report
2. File stored in Azure Blob Storage
3. Sent to GPT-4o Vision for parsing (if image) or GPT-4o for text extraction (if PDF)
4. AI extracts structured health markers: Cholesterol panel (Total, LDL, HDL, Triglycerides), Blood sugar (Fasting Glucose, HbA1c), Complete blood count markers, Vitamin levels (D, B12), Thyroid (TSH, T3, T4), Kidney/liver function markers
5. Parsed data stored as JSONB in health_reports table
6. All future AI Dietitian conversations include this data

6.8 Meal Suggestion Engine Module

Purpose: Proactively suggests meals that optimally fill nutritional gaps.

Algorithm:

1. Calculate remaining nutritional budget for the day
2. Determine expected meal type from time of day
3. Apply dietary filters (vegetarian, allergies, etc.)
4. Apply health condition constraints (low sodium for hypertension, etc.)
5. Request GPT-4o to suggest 3 meals that:
 - Fill the specific caloric and macro gaps
 - Address any micronutrient deficiencies
 - Respect all dietary and health constraints
 - Are culturally appropriate to user preferences
6. Return suggestions with nutritional breakdown

CHAPTER 7

TECHNOLOGY STACK

7.1 Frontend (Mobile Application)

Table 7.1a: Frontend Technologies

Technology	Version	Purpose
React Native	0.73+	Cross-platform mobile framework
Expo	SDK 50+	Development tooling and build system
React Navigation	6.x	Screen navigation and routing
Victory Native / react-native-chart-kit	Latest	Interactive charts and graphs
React Native Gifted Chat	2.x	Chat UI for AI Dietitian
Expo Camera	Latest	Camera access for food photos
Expo Image Picker	Latest	Gallery image selection
Axios	1.x	HTTP client for API communication
AsyncStorage	Latest	Local token and cache storage
React Native Reanimated	3.x	Smooth animations
React Native SVG	Latest	Custom chart elements

7.2 Backend (Server-Side)

Table 7.1b: Backend Technologies

Technology	Version	Purpose
Node.js	20 LTS	JavaScript runtime
Express.js	4.x	REST API framework
bcryptjs	2.x	Password hashing
jsonwebtoken	9.x	JWT token generation & verification
multer	1.x	File upload middleware
pg (node-postgres)	8.x	PostgreSQL client
@azure/storage-blob	12.x	Azure Blob Storage SDK
openai	4.x	Azure OpenAI SDK
cors	2.x	Cross-origin resource sharing
dotenv	16.x	Environment variable management
joi	17.x	Input validation

7.3 Cloud Services (Microsoft Azure)

Table 7.1c: Azure Cloud Services

Service	Tier	Purpose
Azure App Service	B1 Basic	Host Node.js backend API
Azure Database for PostgreSQL Flexible Server	Burstable B1ms	Primary database with pgvector
Azure Blob Storage	Standard LRS	Store food images & health reports
Azure OpenAI Service	Standard S0	GPT-4o + text-embedding-ada-002

7.4 AI/ML Services

Table 7.1d: AI/ML Models

Model	Provider	Purpose
GPT-4o	Azure OpenAI	Food image analysis (Vision), Dietitian chat, Health report parsing
text-embedding-ada-002	Azure OpenAI	1536-dim vector embeddings for similarity matching

7.5 Database

Table 7.1e: Database Technologies

Technology	Version	Purpose
PostgreSQL	16	Relational database
pgvector	0.5+	Vector similarity search extension

CHAPTER 8

DATABASE DESIGN

8.1 Schema Overview

The database consists of 6 tables organized around the user entity, with vector embedding support through the pgvector extension for AI-powered similarity matching.

8.2 Table Descriptions

Table 8.1: Users Table

Column	Data Type	Constraints	Description
id	SERIAL	PRIMARY KEY	Auto-incrementing user ID
email	VARCHAR(255)	UNIQUE, NOT NULL	User email address
password_hash	VARCHAR(255)	NOT NULL	Bcrypt hashed password
name	VARCHAR(100)		User's full name
age	INTEGER		Age in years
gender	VARCHAR(20)		Male/Female/Other
height	DECIMAL(5,2)		Height in cm
weight	DECIMAL(5,2)		Weight in kg
activity_level	VARCHAR(20)		sedentary/light/moderate/active/very_active
goal	VARCHAR(20)		lose/gain/maintain
caloric_target	INTEGER		Daily calorie target (kcal)
protein_target	INTEGER		Daily protein target (g)
carb_target	INTEGER		Daily carbohydrate target (g)
fat_target	INTEGER		Daily fat target (g)
dietary_preferences	TEXT[]		Array: vegetarian, vegan, keto, etc.
allergies	TEXT[]		Array: nuts, dairy, gluten, etc.
health_conditions	TEXT[]		Array: diabetes, hypertension, etc.
embedding	VECTOR(1536)		User profile embedding
created_at	TIMESTAMP	DEFAULT NOW()	Account creation time
updated_at	TIMESTAMP	DEFAULT NOW()	Last profile update

Table 8.2: Meals Table

Column	Data Type	Constraints	Description
id	SERIAL	PRIMARY KEY	Auto-incrementing meal ID
user_id	INTEGER	FK → users(id)	Associated user
meal_type	VARCHAR(20)		breakfast/lunch/dinner/snack
name	VARCHAR(255)		Meal name
image_url	TEXT		Azure Blob Storage URL
logged_at	TIMESTAMP	DEFAULT NOW()	When the meal was logged
is_manual	BOOLEAN	DEFAULT FALSE	Whether manually entered

Table 8.3: Nutrition Entries Table

Column	Data Type	Constraints	Description
id	SERIAL	PRIMARY KEY	Auto-incrementing entry ID
meal_id	INTEGER	FK → meals(id)	Associated meal
food_name	VARCHAR(255)		Individual food item name
calories	DECIMAL(8,2)		Calories (kcal)
protein	DECIMAL(8,2)		Protein (g)
carbohydrates	DECIMAL(8,2)		Carbohydrates (g)
total_fat	DECIMAL(8,2)		Total fat (g)
saturated_fat	DECIMAL(8,2)		Saturated fat (g)
fiber	DECIMAL(8,2)		Dietary fiber (g)
sugar	DECIMAL(8,2)		Sugar (g)
sodium	DECIMAL(8,2)		Sodium (mg)
cholesterol	DECIMAL(8,2)		Cholesterol (mg)
vitamins	JSONB		Vitamin values as % DV
minerals	JSONB		Mineral values as % DV
serving_size	VARCHAR(50)		Serving size description
embedding	VECTOR(1536)		Food item embedding

Table 8.4: Health Reports Table

Column	Data Type	Constraints	Description
id	SERIAL	PRIMARY KEY	Auto-incrementing report ID
user_id	INTEGER	FK → users(id)	Associated user
report_type	VARCHAR(50)		blood_work, general_checkup, etc.
file_url	TEXT		Azure Blob Storage URL
parsed_data	JSONB		AI-extracted health markers
uploaded_at	TIMESTAMP	DEFAULT NOW()	Upload timestamp

Table 8.5: Chat Messages Table

Column	Data Type	Constraints	Description
id	SERIAL	PRIMARY KEY	Auto-incrementing message ID
user_id	INTEGER	FK → users(id)	Associated user
role	VARCHAR(20)		user / assistant
content	TEXT		Message text
created_at	TIMESTAMP	DEFAULT NOW()	Message timestamp

Table 8.6: Daily Summaries Table

Column	Data Type	Constraints	Description
id	SERIAL	PRIMARY KEY	Auto-incrementing summary ID
user_id	INTEGER	FK → users(id)	Associated user
date	DATE	NOT NULL	Summary date
total_calories	DECIMAL(8,2)		Aggregate daily calories
total_protein	DECIMAL(8,2)		Aggregate daily protein
total_carbs	DECIMAL(8,2)		Aggregate daily carbs
total_fat	DECIMAL(8,2)		Aggregate daily fat
total_fiber	DECIMAL(8,2)		Aggregate daily fiber
meal_count	INTEGER		Number of meals logged
<i>UNIQUE constraint on (user_id, date) — one summary per user per day</i>			

8.3 Relationships

- **Users → Meals:** One-to-Many (a user logs multiple meals)
- **Meals → Nutrition Entries:** One-to-Many (a meal contains multiple food items)
- **Users → Health Reports:** One-to-Many (a user uploads multiple reports)
- **Users → Chat Messages:** One-to-Many (a user sends multiple messages)
- **Users → Daily Summaries:** One-to-Many (one summary per day per user)

CHAPTER 9

SYSTEM REQUIREMENTS

9.1 Hardware Requirements

Table 9.1: Hardware Requirements

Component	Development	Deployment (Azure)	End User
Processor	Intel i5 / AMD Ryzen 5 or above	1 vCPU (B1ms)	ARM/x86 mobile processor
RAM	8 GB minimum, 16 GB recommended	2 GB (B1ms)	4 GB minimum
Storage	50 GB free disk space	32 GB SSD (Azure)	200 MB app + camera storage
Display	1920×1080 recommended	N/A	720p or higher Android display
Network	Broadband internet	Azure network	4G / Wi-Fi connectivity
Camera	N/A	N/A	8 MP or higher (for food photos)

9.2 Software Requirements

Table 9.2: Software Requirements

Component	Requirement
Development Machine	
Operating System	Windows 10/11, macOS 12+, or Ubuntu 20.04+
Node.js	v20.x LTS
npm	v10.x
Android Studio	Hedgehog (2023.1+) with Android SDK
Expo CLI	Latest
VS Code	Latest (recommended IDE)
Git	2.40+
Deployment (Azure)	
Azure App Service Runtime	Node.js 20 LTS
PostgreSQL	16 with pgvector 0.5+
Azure OpenAI	GPT-4o deployment, ada-002 deployment
End User Device	
Android Version	Android 10 (API 29) or higher
Expo Go (for testing)	Latest from Google Play Store

CHAPTER 10

IMPLEMENTATION PLAN & TIMELINE

Table 10.1: Phased Implementation Schedule

Phase	Duration	Tasks	Deliverables
Phase 1: Setup & Foundation	Week 1–2	Azure account setup; Provision PostgreSQL + Blob Storage + OpenAI; Initialize React Native project; Backend Express.js boilerplate	Working dev environment, empty project structure, Azure services running
Phase 2: Authentication & Profile	Week 3–4	User registration & login API (bcrypt + JWT); Profile setup screen; Health profile API; Database schema deployment	Functional auth flow, profile creation & storage
Phase 3: Food Analysis Core	Week 5–7	Food image capture UI; GPT-4o Vision integration; Nutrition extraction & parsing; Meal storage API; Analysis results display	Working food photo → nutrition analysis pipeline
Phase 4: Daily Tracking & Dashboard	Week 8–9	Today's meals API; Dashboard screen with progress indicators; Daily summaries computation; Meal history API & screen	Complete meal tracking with daily view
Phase 5: Charts & Analytics	Week 10–11	Calorie trend line chart; Macro donut chart; Weekly bar chart; Nutrient radar chart; Trends API	Full analytics dashboard with 4 chart types
Phase 6: AI Dietitian Chat	Week 12–14	Chat UI; Context builder service; GPT-4o chat integration; Chat history storage; Proactive meal suggestion API	Working conversational AI dietitian
Phase 7: Health Reports	Week 15–16	Report upload UI & API; GPT-4o document parsing; Health marker extraction & storage; Integration with chat context	Health report upload → parsed data → dietary integration
Phase 8: Polish & Testing	Week 17–18	UI animations & transitions; Error handling; Edge case testing; Performance optimization; User acceptance testing	Production-quality app
Phase 9: Deployment & Documentation	Week 19–20	Deploy backend to Azure App Service; Build signed APK; Prepare project report; Create demo video	Final APK, deployed backend, complete documentation

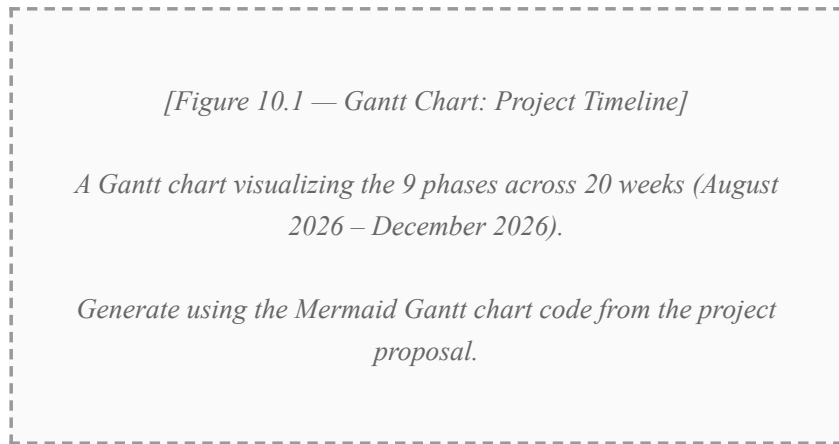


Figure 10.1: Gantt Chart — Project Implementation Timeline

CHAPTER 11

EXPECTED OUTCOMES

Upon successful completion of this project, the following outcomes are expected:

1. **A fully functional Android application (APK)** that allows users to photograph meals and receive instant AI-powered nutritional analysis, achieving at least 85% accuracy in food identification for common Indian and international dishes.
2. **An operational AI Dietitian** capable of providing personalized, context-aware dietary recommendations that consider the user's complete health profile, current daily intake, and health report data — simulating the advisory quality of a professional dietitian consultation.
3. **Interactive nutritional dashboards** with four types of charts (line, donut, bar, radar) that transform raw nutritional data into actionable visual insights, enabling users to understand their dietary patterns at a glance.
4. **A scalable cloud-native backend** deployed on Microsoft Azure, demonstrating proficiency in cloud computing services including App Service, managed databases, blob storage, and AI-as-a-service.
5. **A comprehensive daily meal tracking system** with persistent logging, daily summaries, and historical data retrieval — providing the foundation for long-term dietary adherence monitoring.
6. **Health report parsing capability** that bridges the gap between clinical health data and daily dietary choices, making health report data actionable in real-time meal decisions.
7. **A reusable project architecture** that can serve as a template for future health-tech applications combining mobile development, cloud computing, and AI services.

CHAPTER 12

FUTURE SCOPE

The following enhancements can be pursued to extend MealMentor AI beyond the current project scope:

1. **iOS Application:** Leverage React Native's cross-platform capability to build and publish an iOS version on the Apple App Store with minimal additional development.
2. **Barcode & Label Scanning:** Integrate barcode scanning for packaged foods, linking to nutritional databases (Open Food Facts API) for precise pre-packaged food tracking.
3. **Wearable Device Integration:** Connect with smartwatches and fitness bands (Google Fit, Samsung Health) to incorporate real-time activity data (steps, heart rate, calories burned) into dietary recommendations.
4. **Meal Plan Generation:** Extend the AI Dietitian to generate complete weekly meal plans with grocery shopping lists, recipes, and preparation instructions.
5. **Multi-Language Support:** Add support for Hindi, Marathi, Tamil, and other Indian regional languages to increase accessibility.
6. **Social Features:** Enable users to share meals, follow friends' nutrition journeys, and participate in dietary challenges for motivation.
7. **Restaurant Menu Integration:** Partner with food delivery platforms (Zomato, Swiggy) to analyze restaurant menu items and suggest the healthiest options from any menu.
8. **Predictive Health Alerts:** Use longitudinal dietary data and health reports to predict potential health risks (e.g., "Your sodium intake trend may increase blood pressure risk") and suggest preventive dietary changes.
9. **Offline AI Mode:** Implement on-device AI inference using TensorFlow Lite for basic food recognition when internet connectivity is unavailable.
10. **Professional Dietitian Dashboard:** Create a web portal where certified dietitians can review patients' MealMentor AI data, approve/modify AI recommendations, and conduct virtual consultations.

CHAPTER 13

REFERENCES

- [1] Mezgec, S., & Koroušić Seljak, B. (2017). NutriNet: A Deep Learning Food and Drink Image Recognition System for Dietary Assessment. *Nutrients*, 9(7), 657. <https://doi.org/10.3390/nu9070657>
- [2] Bossard, L., Guillaumin, M., & Van Gool, L. (2014). Food-101 – Mining Discriminative Components with Random Forests. *European Conference on Computer Vision (ECCV)*, 446–461.
- [3] Sahoo, D., Hao, W., Ke, S., et al. (2019). FoodAI: Food Image Recognition via Deep Learning for Smart Food Logging. *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*.
- [4] Chen, J., Zhu, B., Ngo, C.W., Chua, T.S., & Jiang, Y.G. (2020). A Study of Multi-Task and Region-Wise Deep Learning for Food Ingredient Recognition. *IEEE Transactions on Image Processing*, 30, 1514–1526.
- [5] Vaswani, A., Shazeer, N., Parmar, N., et al. (2017). Attention Is All You Need. *Advances in Neural Information Processing Systems (NeurIPS)*, 30.
- [6] Achiam, J., Adler, S., Agarwal, S., et al. (2023). GPT-4 Technical Report. *OpenAI*. <https://arxiv.org/abs/2303.08774>
- [7] Mifflin, M.D., St Jeor, S.T., Hill, L.A., Scott, B.J., Daugherty, S.A., & Koh, Y.O. (1990). A new predictive equation for resting energy expenditure in healthy individuals. *American Journal of Clinical Nutrition*, 51(2), 241–247.
- [8] World Health Organization. (2021). *Noncommunicable diseases: Key facts*. <https://www.who.int/news-room/fact-sheets/detail/noncommunicable-diseases>
- [9] International Institute for Population Sciences (IIPS). (2021). *National Family Health Survey (NFHS-5), 2019-2021: India Report*. Ministry of Health and Family Welfare, Government of India.
- [10] Microsoft Azure Documentation. (2024). *Azure OpenAI Service*. <https://learn.microsoft.com/en-us/azure/ai-services/openai/>
- [11] Microsoft Azure Documentation. (2024). *Azure Database for PostgreSQL - Flexible Server*. <https://learn.microsoft.com/en-us/azure/postgresql/>
- [12] Microsoft Azure Documentation. (2024). *Azure Blob Storage*. <https://learn.microsoft.com/en-us/azure/storage/blobs/>
- [13] React Native Documentation. (2024). *Getting Started*. <https://reactnative.dev/docs/getting-started>
- [14] Expo Documentation. (2024). *Introduction to Expo*. <https://docs.expo.dev/>

- [15] pgvector. (2024). *Open-source vector similarity search for Postgres*.
<https://github.com/pgvector/pgvector>